Auroral Records and Solar Cycle

1 Point-process Periodogram

$$I(f) = \frac{1}{N}\left|\sum_{n=1}^{N} e^{2\pi i f t_n}\right|^2 = \frac{1}{N}\left[\left(\sum \cos 2\pi f t_n\right)^2 + \left(\sum \sin 2\pi f t_n\right)^2\right]$$

This is the [Bartlett, 1963] spectral estimate for point processes. It determines the frequency f at which the event times t_n align coherently in phase. The estimator requires no assumption about the form of the intensity function $\lambda(t)$ and is asymptotically consistent as $N \to \infty$. It answers the question:

> *For which candidate period do the observed auroral event times show the strongest phase coherence?*

With $T = 870$ years and $P \approx 10$ years there are approximately 87 observable cycles, giving a period of

$$\Delta P \approx \frac{P^2}{T} \approx 0.11 \text{ years},$$

Therefore, I expect the peaks across the range $[7, 16]$ years. The output is put in the next section to compare with Modified Least-Squares method

Result: Dominant peak at $P = 9.937$ years.

2 Modified Least-Squares

The Modified Least-Squares method from [Quinn et al., 2012] estimates the period by finding the candidate frequency that gives the smallest phase misalignment between the observed event times. It is defined as

$$\mathrm{SSMOD}(\gamma) = \min_{\nu} \sum_{n=1}^{N} \langle \gamma t_n - \nu \rangle^2 ,$$

where $\gamma = 1/P$ is the candidate frequency, ν is the phase offset, and

$$\langle x \rangle = x - \mathrm{round}(x).$$

The wrapping operation is used because the phase is circular. Therefore, events near the end of one cycle and the beginning of the next cycle are still considered close to each other. The method answers the question:

For which candidate period do the observed auroral events align most closely within a repeating cycle?

For each candidate period between 7 and 16 years, I calculate the phase misalignment and search for the minimum value of SSMOD. In contrast to the Point-process Periodogram, where a higher value represents stronger phase coherence, a lower SSMOD value represents better phase alignment.

In the implementation, I use 800 candidate frequencies and 400 possible phase offsets for each frequency.

Result: The minimum occurs at $P = 9.919$ years.

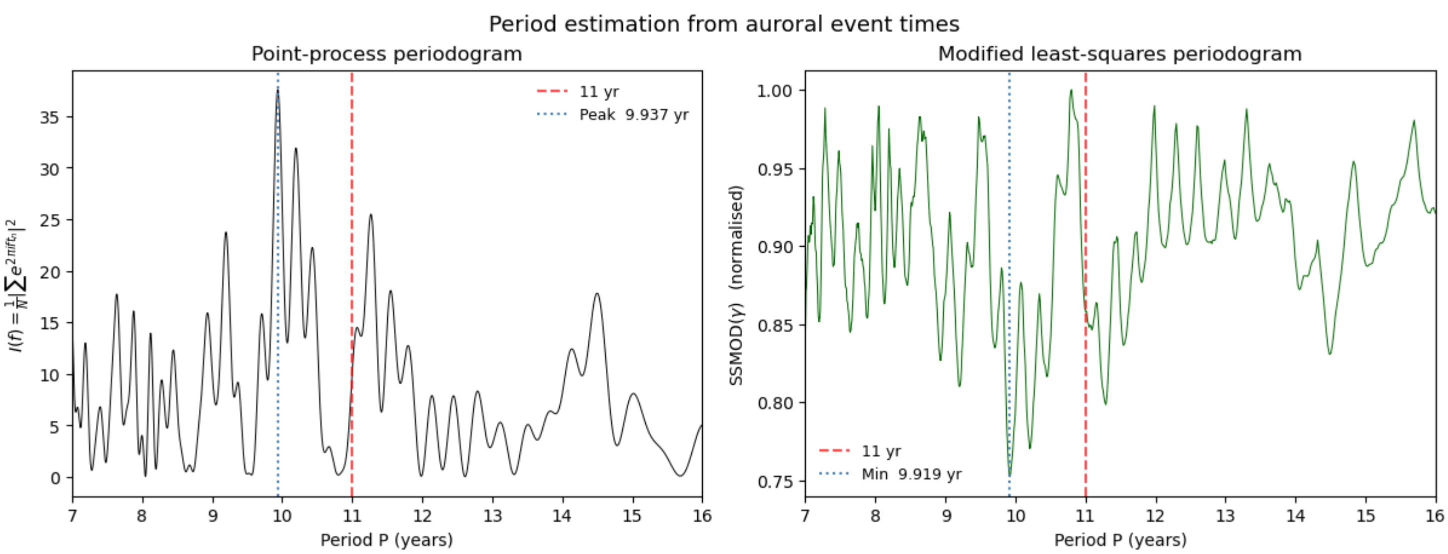

Figure 1: Comparison of the Point-process Periodogram and Modified Least-Squares results. The Point-process Periodogram has its dominant peak at $P = 9.9375$ years, while Modified Least-Squares has its minimum at $P = 9.919$ years.

The two methods look at the event times in different ways. The Point-process Periodogram searches for the period with the strongest phase coherence, while Modified Least-Squares searches for the period with the smallest phase misalignment. However, they give very similar estimates:

$$P_{\mathrm{PP}} = 9.9375 \text{ years}, \qquad P_{\mathrm{LLS}} = 9.919 \text{ years}.$$

Therefore, both methods identify a period close to 9.9 years in the Korean auroral records.

3 Sequential Period Recovery

After applying the Point-process Periodogram to the full dataset, I also want to examine how the estimated period changes as more auroral events become available. This answers the question:

> *How much historical data is needed before the Point-process Periodogram begins to recover a stable period?*

To examine this, the auroral events are added chronologically in batches of five and the Point-process Periodogram is recalculated after each batch.

The animation contains two panels. The top panel shows the auroral events that have been included up to the current point in time. The highlighted region marks the burst period from 1535 to 1563 CE.

I the frame shown below, only 10 events have been observed. At this stage, the dominant period is estimated as approximately 12.04 years. As additional events are included, the shape of the periodogram and the location of the dominant peak continue to change.

Therefore, similar to the previous NHPP analysis, this animation is used to examine when the estimated period begins to stabilise as more events are observed. In this case, the estimate eventually approaches the full-data result of $P \approx 9.94$ years.

References

[Bartlett, 1963] Bartlett, M. S. (1963). The spectral analysis of point processes. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 25(2):264–281.

[Quinn et al., 2012] Quinn, B. G., Clarkson, I. V. L., and McKilliam, R. G. (2012). Estimating period from sparse, noisy timing data. In *2012 IEEE Statistical Signal Processing Workshop (SSP)*, pages 193–196. IEEE.